

FED3 Fiber Photometry

Processing, post-processing and FiPhoPHA

Code-checked user and methods guide | Review edition 1.0 | 2 September 2026

Prepared from the supplied FED3 guide, ClassicFED3_Fiber_Photometry_Arduino.txt and the local Andrews Lab photometry GUI code. This guide describes the implementation reviewed on this date. It does not modify the Arduino sketch, TDT configuration or analysis code.

The essential rule: verify the behavioural meaning of the selected TDT store before interpreting any sheet called Left, Right, Rewarded or Pellet. A store name or GUI default is not a universal behavioural definition.

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1. Scope and workflow

This guide is specifically for FED3-associated fiber photometry. It covers recording provenance, Arduino/BNC/TDT event interpretation, the FED3 processing branch of the GUI, FED3 post-processing and the local FED3-aware FiPhoPHA between-group analysis. It does not teach the unrelated Peri-events, Whole recording or Between events workflows.

The workflow has three analysis stages. First, the GUI extracts peri-event fluorescence around selected TDT onsets and exports one workbook per signal type and recording/setup. Second, FED3 post-processing combines compatible z-score workbooks and calculates event metrics and descriptive plots. Third, FED3 FiPhoPHA reads the combined workbook and performs bootstrap and permutation analyses. The FED CSV is useful for validating event identities but is not automatically imported into these three analysis stages.

1.1 Quick-start checklist

- Preserve the complete TDT block, FED CSV, Arduino sketch, FED3 library version and TDT/Synapse configuration used for the session.
- Confirm the physical active side, session mode, BNC routing and pulse-to-store decoding. Record a mapping for each mode, not merely for each rig.
- Process each animal/setup with the correct signal/control streams, event mapping, peri-event range and normalization settings.
- Export zScore workbooks for the FED3 post-processing route. Keep settings and original outputs unchanged in an archive.
- Combine only compatible sessions and event meanings. Supply unique file basenames and unique mouse identifiers for each combined run.
- Check FED3_FP_Combined.xlsx, its Analysis Parameters sheet, event counts and plots before statistics.
- Choose the FiPhoPHA analysis unit, grouping, trace range, windows and temporal threshold deliberately. Save the seed and all settings.

1.2 Files to keep together

Use a project folder with separate locations for raw TDT blocks, raw FED logs, processing outputs, combined outputs and statistical outputs. The GUI saves many outputs under fixed names, so separate run folders prevent accidental replacement of earlier analyses.

A reproducible release should include this guide, ClassicFED3_Fiber_Photometry_Arduino.txt, the matching TDT/Synapse configuration or circuit screenshots, and a validated mapping table. The Arduino file alone is insufficient: it specifies electrical patterns, not the names TDT assigns to them.

Record the software revision, dependencies, sensor, acquisition sampling/filter settings, animal, session, physical active port, setup, camera, source stores, GUI category mapping and event counts. A setup identifier does not establish an animal identifier.

2. Understanding FED3, BNC and TDT events

2.1 Four different meanings must stay separate

Layer	What it describes	Example
Behaviour	Physical port and behavioural outcome	Physical right poke earns a pellet in FR1R
Arduino output	Electrical pattern sent by the selected mode	BNC(50, 1)
TDT store	Name assigned by recording/decoder configuration	Alft
GUI category	Output label selected by the user	Left sheet, if its dropdown selects Alft

The A and B prefixes identify the recording setup in the lab convention: Alft/Blft, Argt/Brgt, Arwd/Brwd and Aplt/Bplt. They do not mean active/inactive. The suffixes are names, not proof of physical side or reward status. BNC wiring determines which input receives a signal; Arduino timing and TDT decoding determine how it is classified.

The GUI prefers matching setup-specific names: Alft/ALft, Argt/ARgt, Aplt/APlt and Arwd/ARwd for A, and corresponding B names for B. If absent, it tries generic Left, Rght/RGht, Pelt and Rewd names. If no unique match is found, it can fall back to the first available store. This fallback is not a validated behavioural choice.

The dropdowns remain user-selectable. Selecting Left = Argt puts Argt onsets in the Left sheet; it does not change the acquisition, decode a different pulse or automatically relabel the physical port. Never swap dropdowns merely because a session is reversed without checking its recorded event mapping.

2.2 What BNC(d, n) means

In the FED3 library implementation inspected, BNC(d, n) sends n cycles of d milliseconds HIGH followed by d milliseconds LOW. Thus the repetition frequency is $1000/(2d)$ Hz and the call takes approximately $2dn$ milliseconds, including the final LOW interval. A single pulse has a width, not a sustained train frequency. The exact compiled FED3 library should be archived with the sketch.

Call	Output	Approximate call duration
BNC(45, 1)	One 45-ms HIGH / 45-ms LOW cycle	90 ms
BNC(45, 3)	Three such cycles, about 11.1 Hz	270 ms

Call	Output	Approximate call duration
BNC(30, 5)	Five 30-ms HIGH / 30-ms LOW cycles, about 16.7 Hz	300 ms
BNC(50, 1), (50, 2), (50, 3)	One, two or three 50-ms HIGH / 50-ms LOW cycles	100, 200 or 300 ms
BNC(25, 20)	Twenty 25-ms HIGH / 25-ms LOW cycles, 20 Hz	1 second

Two one-pulse calls on the same rewarded trial can form a two-pulse code if the TDT decoder groups them. They are not necessarily two separate recorded poke events. Conversely, knowing pulse count alone does not prove which epoc name is created without inspecting the decoder.

2.3 The supplied sketch: mode-specific outputs

The following table reports explicit calls in the supplied sketch. 'After Feed' means after that library function returns, not necessarily immediately after dispensing. Other library-internal behaviour is not fully specified by the sketch itself.

Mode	Poke/output rule	Reward-related sequence
0: Free feeding	No poke requirement in this branch	Feed(); then BNC(50,1)
1-3: FR1, FR3, FR5	Left: BNC(45,1). Right: BNC(30,5)	Successful left: an additional BNC(45,1), cue, Feed(), then BNC(45,3)
4: PR	Left: BNC(45,1), Click(), increment counter. Right: BNC(30,5)	Ratio completion: cue, additional BNC(45,1), Feed(), BNC(45,3), update ratio/reset counter
5: Extinction	Left: BNC(50,1) plus cue. Right: BNC(30,5)	No Feed() call in this branch
6: Light tracking	Active side selected by activePoke; successful response cues and feeds, then randomizes side	No explicit BNC call in this branch
7: FR1 reversed	Physical left: BNC(30,5). Physical right: BNC(50,1)	Every handled right response cues and feeds, then BNC(50,3)
8: PR reversed	Physical right: BNC(50,3), increment counter. Physical left: BNC(50,2)	Ratio completion: cue, Feed(), then BNC(50,1)
9/10: Optogenetic	Mode 9 left / mode 10 right: cue and BNC(25,20)	No pellet delivery in these branches

Mode	Poke/output rule	Reward-related sequence
11: Timed FR1	Left feeds only during configured hours; right is logged	No explicit BNC call in this branch

For ordinary PR, the active counter starts at zero and the first requirement is one. After a pellet, the next requirement is $\text{round}(5 \exp(0.2(\text{PelletCount} + 1)) - 5)$; fed3.FR is updated and poke_num resets to zero. These are handled active responses in the PR loop, not every possible CSV event such as LeftDuringDispense. The fixed-ratio branch instead tests LeftCount modulo FR.

IMPORTANT: FR1R does not emit the extra pre-Feed pulse used by normal FR/PR. A one-pulse Alft-like classification can therefore represent a rewarded physical-right response in FR1R. Reversed PR uses yet another pulse assignment. Do not apply the normal-PR interpretation to these modes automatically.

2.4 Rewarded poke versus pellet retrieval

Feed() may run the motor and wait for pellet removal before returning. A BNC call placed after Feed() can therefore mark retrieval rather than dispensing. LogLeftPoke()/logRightPoke() and other library functions may also delay the explicit BNC call relative to initial nose entry. 'Event onset' in the exported photometry is a TDT event onset, not a guaranteed sensor beam-break instant.

The supplied sketch's post-Feed three-pulse call is consistent with Aplt/Bplt representing pellet retrieval in the normal PR example. But this is not universal: Mode 0 sends one pulse after Feed(), and reversed PR also sends one pulse after Feed(). Use the actual compiled library and decoder to establish timing for each mode.

2.5 Evidence from the reviewed PR example

In one PR FED3 example, the FED CSV contained 64 ordinary Left events, one LeftDuringDispense, 16 Right events and seven Pellet events. The TDT block contained 57 Alft, seven Arwd, 16 Argt and seven Aplt onsets. The relationship $64 = 57 + 7$, together with the sketch's one-/two-pulse pattern, strongly supports Alft as unrewarded active and Arwd as rewarded active for that configuration. The separate dispensing-time poke is not represented in that sum.

For this normal-PR mapping, the existing Left sheet already contains unrewarded active events. A conceptual 'Unrewarded all' category would combine Alft and Argt, not subtract Arwd again. Such a combined category is not implemented in the reviewed processing/post-processing menus. It cannot be inferred safely for FR1R or reversed PR from the normal-PR rule.

3. Processing a FED3 recording

Source: Run_program.py; GUI_and_data_processing/Create_GUI.py; Analysis_types/FED3.py.

3.1 Select the block, setup and streams

Launch the installed Andrews Lab Photometry Analysis GUI using the repository's installation instructions. Start without an existing settings workbook for a first manual run. Select the complete TDT block folder as the import location and a dedicated output folder. Keep the .tsq, .tev and associated block files together. Select FED3 as the analysis type.

Choose the setup and matching camera for the animal. Camera selection is especially important for video snippets; it does not automatically establish the signal or behavioural mapping.

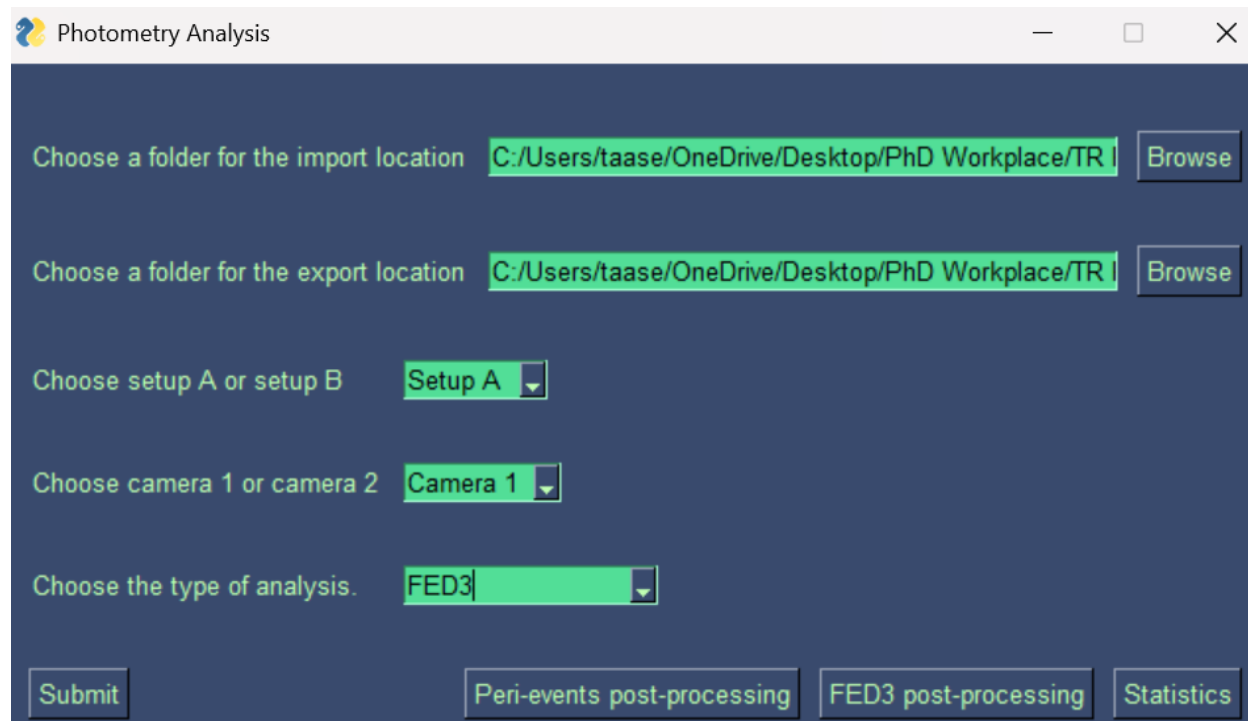
The image shows a screenshot of a software window titled "Photometry Analysis". The window has a dark blue background with white text and green highlights for the input fields. It contains several configuration options: "Choose a folder for the import location" with a text field containing "C:/Users/taase/OneDrive/Desktop/PhD Workplace/TR" and a "Browse" button; "Choose a folder for the export location" with a text field containing "C:/Users/taase/OneDrive/Desktop/PhD Workplace/TR" and a "Browse" button; "Choose setup A or setup B" with a dropdown menu showing "Setup A"; "Choose camera 1 or camera 2" with a dropdown menu showing "Camera 1"; and "Choose the type of analysis." with a dropdown menu showing "FED3". At the bottom, there are four buttons: "Submit", "Peri-events post-processing", "FED3 post-processing", and "Statistics".

Figure 1. Main processing window: select the block, output folder, setup, camera and FED3 analysis.

The next window proposes control and signal streams based on the setup. Typical A options are _405A for ISOS and _465A for signal; B supports _415A or _405C for ISOS and _475A or _465C for signal. Some alternative sensor channels are also supported. TDT's numeric-leading store names may appear with a leading underscore in Python. Verify acquisition records rather than accepting an automatically chosen channel simply because it is available.

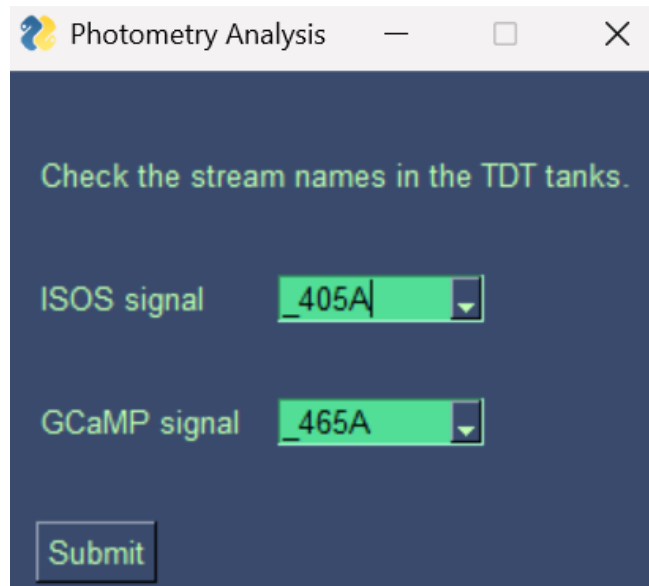


Figure 2. Signal/control selection. The GCaMP label is the GUI's signal-field name; use the stream appropriate to the actual sensor.

3.2 Choose task and map events

Select Normal task for the usual FED3 workflow. Pavlovian task has Stimulus and Pellet categories, not the four normal categories; the downstream FED3 tools reviewed here only recognize Left, Right, Pellet and Rewarded. Do not assume the full downstream route supports the Pavlovian Stimulus category.

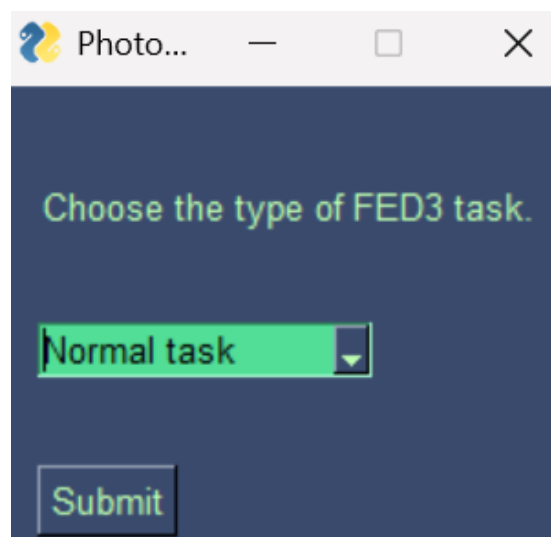
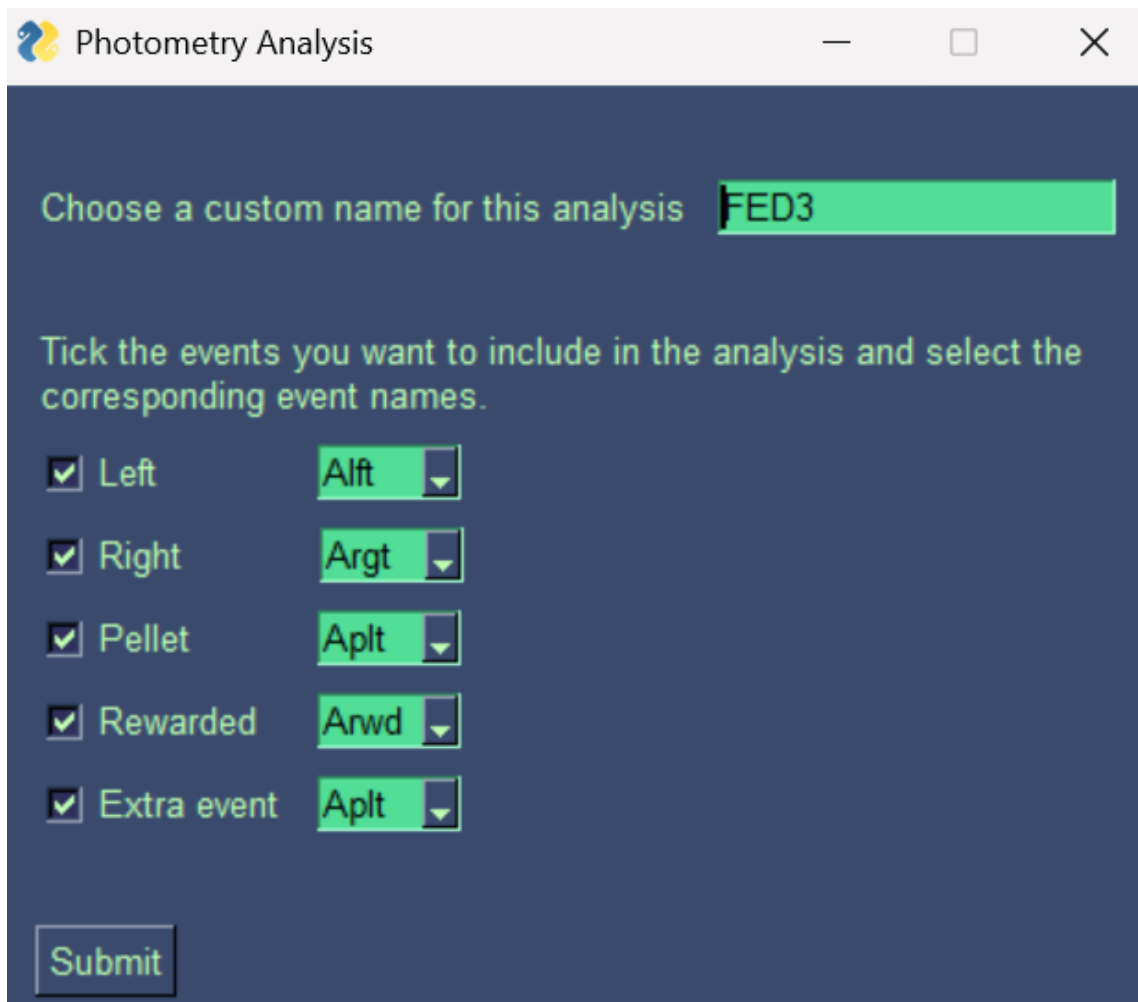


Figure 3. FED3 task-type selection.

Give the analysis a meaningful name and tick only the categories required. For each, select the verified TDT store. Leave Extra event unticked unless its inclusion is intentional. Check all mappings again when changing session mode or rig.



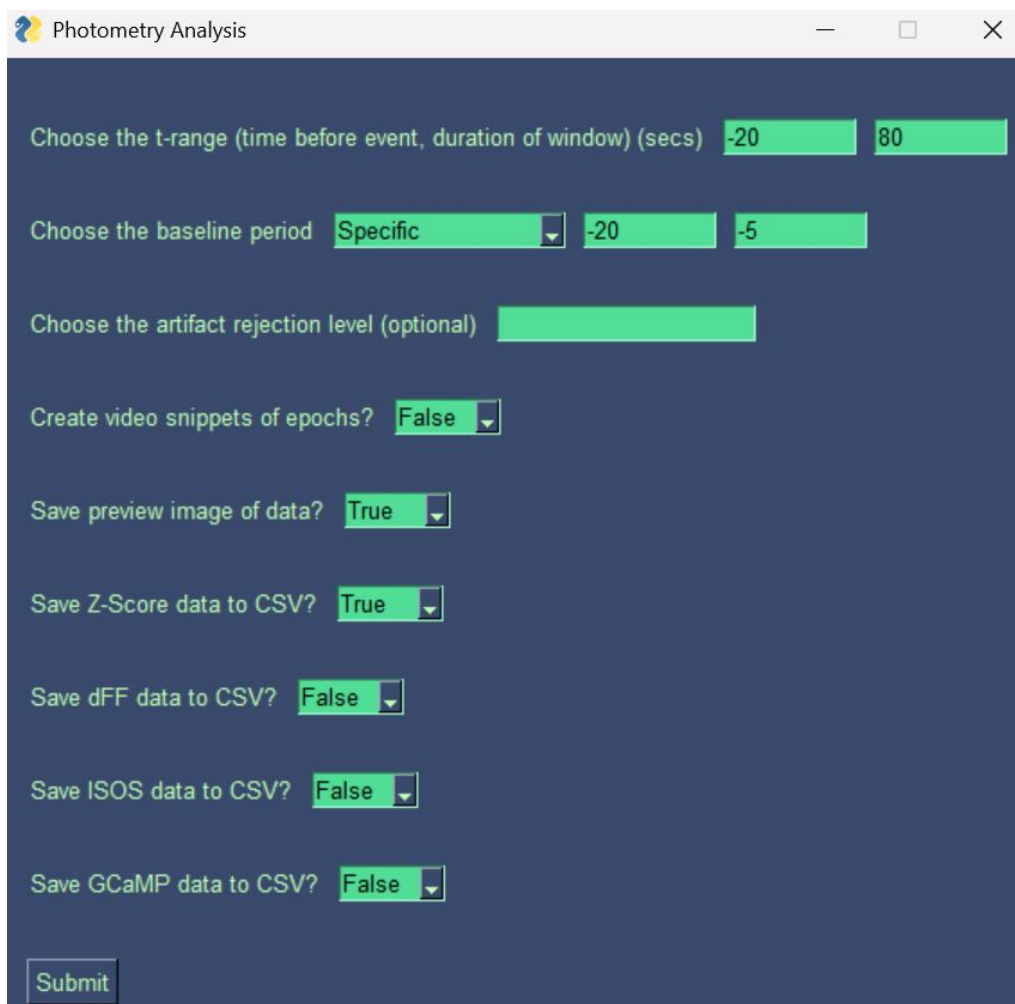
The screenshot shows a window titled "Photometry Analysis" with a standard macOS-style title bar. The main content area has a dark blue background. At the top, there is a text prompt "Choose a custom name for this analysis" followed by a text input field containing "FED3". Below this, another text prompt reads "Tick the events you want to include in the analysis and select the corresponding event names." There are five rows of event selection options, each consisting of a checked checkbox, an event name, and a dropdown menu. The options are: "Left" with "Alft", "Right" with "Argt", "Pellet" with "Aplt", "Rewarded" with "Arwd", and "Extra event" with "Aplt". A "Submit" button is located at the bottom left of the form.

Event	Selected Name
☒ Left	Alft
☒ Right	Argt
☒ Pellet	Aplt
☒ Rewarded	Arwd
☒ Extra event	Aplt

Figure 4. Event mapping example. Extra event is checked and repeats Aplt in this screenshot; uncheck it if that duplication is unintended.

If the same store is selected twice, the GUI copies its onsets twice under different notes. Those duplicates enter Overall and change its weighting. Selecting a B-prefixed store in an A analysis can include the other rig's events. These are not automatically prevented.

3.3 Peri-event settings



The screenshot shows the 'Photometry Analysis' window with the following settings:

- Choose the t-range (time before event, duration of window) (secs):** -20, 80
- Choose the baseline period:** Specific, -20, -5
- Choose the artifact rejection level (optional):** (empty field)
- Create video snippets of epochs?** False
- Save preview image of data?** True
- Save Z-Score data to CSV?** True
- Save dFF data to CSV?** False
- Save ISOS data to CSV?** False
- Save GCaMP data to CSV?** False
- Submit** button

Figure 5. Peri-event options. Defaults are starting values, not a universal experimental protocol.

Setting	Meaning in the reviewed code
t-range	Start relative to the event, then duration, both in seconds. Default [-20,80] means approximately -20 to +60 s, not -20 to +80 s.
Specific baseline	Default -20 to -5 s. Samples strictly inside these bounds define each trial's residual mean and SD.
Whole recording baseline	Uses mean/SD of the separately fitted full-recording residual. Not identical to normalizing each trial with its own baseline.
Artifact rejection level	Blank becomes infinity. Finite-threshold behaviour has implementation caveats; see Section 4.4.
Video snippets	Default False. Requires the correct camera video/timestamps; this is a QC aid, not an event reclassifier.

Setting	Meaning in the reviewed code
Preview image	Default True. An overview of all selected events, not an independent statistical result.
Signal exports	Default zScore=True; dFF, ISOS and GCaMP=False. Despite 'CSV' in the shared dialog, the FED3 branch writes .xlsx workbooks.

Choose time windows before examining the biological result. Use matching windows, normalization and sampling settings across files intended for combination. Overlapping peri-event windows are not removed: closely spaced pokes can share fluorescence samples, and later reward/retrieval responses can enter an earlier poke's window.

Repeat for Setup B where appropriate. A manually completed run also writes a Settings workbook for reuse. Keep the saved mapping with the output. When loading settings for other recordings, recheck paths, stores, setup and event meanings rather than assuming the original mapping still applies.

4. Processing mathematics and first-stage outputs

Source: Root_Morales_lab_codes/FibPhoEpocAveraging.py; Analysis_types/Peri_events.py; Analysis_types/FED3.py.

4.1 Event assembly and sampling

For every selected GUI category, the code copies all onsets from its chosen epoc and attaches that category as a note. It sorts the combined onset list, assigns numeric event IDs and creates Analyse_this_event. The standard fixed-duration peri-event extraction then uses tdt.epoc_filter. The GUI does not compute reward status by comparing poke and pellet times.

Initial processing sets $N=100$. Nominal output spacing is N/fs , where fs is the stream sampling frequency; therefore this is not necessarily exactly 10 Hz. Event traces are trimmed using $x[1:min_length]$. The averaging loop then steps by N but uses the slice $x[i:i+N-1]$. In Python this contains up to $N-1$ samples, not N . Thus, for $N=100$, normal full blocks average 99 samples while advancing 100. This is the implementation as reviewed, not a recommendation. Short/empty tail slices can also require checking for NaNs.

The reported relative time vector is $t_k = t_start + kN/fs$ for $k=1,...,K$. The first reported time is therefore one nominal step after t_start , not necessarily exactly the requested start. Boundary windows that cannot be extracted are excluded; check_for_excluded_data attempts to synchronize their metadata with the retained traces.

4.2 Control fitting, residual, dFF and z-score

Let $x_i(t)$ be the averaged control trace and $y_i(t)$ the averaged signal trace for event i . A separate ordinary least-squares line is fitted for every event across its extracted window:

$$(a_i, b_i) = \operatorname{argmin}_{a, b} \sum_t [y_i(t) - (a_i x_i(t) + b_i)]^2$$

$$\operatorname{fit}_i(t) = a_i x_i(t) + b_i$$

$$dF_i(t) = y_i(t) - \operatorname{fit}_i(t)$$

$$dFF_i(t) = dF_i(t) / \operatorname{fit}_i(t)$$

dFF is a fraction in this implementation, not multiplied by 100. The z-score is calculated from dF, not by z-scoring the exported dFF array. The control fit is not an explicit exponential bleaching model or a separately implemented low-pass filter.

For Specific baseline, with $B = \{t: \text{baseline_start} < t < \text{baseline_end}\}$:

$$\mu_i = \operatorname{mean}_B(dF_i); \sigma_i = \sqrt{\operatorname{mean}_B((dF_i - \mu_i)^2)}$$

$$z_i(t) = (dF_i(t) - \mu_i) / \sigma_i$$

The SD uses NumPy's population convention ($\text{ddof}=0$). A zero SD or empty baseline is not a meaningful normalization and can yield non-finite results. Verify adequate finite baseline samples and nonzero variation.

Whole recording computes a separate control-to-signal fit on the averaged full recording, obtaining dF_{full} . Its mean and population SD replace μ_i and σ_i in the formula above. The numerator still contains the event-specific residual dF_i . A common normalization reference is used, but the event residual and full-recording residual arise from different fits.

ISOS and GCaMP exports contain the averaged control and signal event arrays. The raw-trace preview subtracts their mean DC offsets for display; this display offset is not applied to the exported raw arrays.

4.3 First-stage workbook and sheet reference

Each enabled signal produces a file of the form `block_signal_analysis_Setup_A.xlsx` (or `Setup_B`), where signal is zScore, dFF, ISOS or GCaMP. The normal FED3 workbook contains Overall and each included GUI category. An optional preview PNG uses the block, analysis name and setup in its name.

Sheet	Contents
Overall	All selected, retained event columns in combined onset order, including any intentionally or accidentally duplicated mappings. Its mean pools event types.
Left	Columns whose Event note is Left; behavioural meaning follows the selected store and task.
Right	Columns whose Event note is Right. It is not automatically all unrewarded behaviour.
Rewarded	Columns whose Event note is Rewarded. The name does not verify the source store.
Pellet	Columns whose Event note is Pellet; often retrieval in the validated normal-PR configuration.
Extra event	Any separately selected source under this label. No predefined behavioural meaning.

The splitter retains the three leading label/time/mean columns and selects trial columns by exact note equality. It recalculates each category's mean across that sheet's event columns. Signal processing is not rerun separately for each sheet. Event IDs can have gaps within a category because IDs originate in the combined list.

Header rows preserve event number, absolute/block-relative onset seconds, Event note, Preceding note, per-trial metrics, source filename and Custom name. Preceding note reports Same/Different relative to the preceding event in the combined sequence; it is not recalculated within each exported category and is not a reward indicator.

Max values is the maximum across the complete extracted trace, including pre-event samples; Time of max values is the first sample time attaining that maximum. 'Max value above 1?' is a numerical flag, not a statistical significance test. The post-processing PeakValues metric is recalculated within a separately chosen window and may differ.

For zScore only, Time to baseline is the first $t \geq 0$ satisfying $-0.1 \leq z_i(t) \leq 0.1$. It can be zero/near-zero even before a later response peak. It is not a sustained recovery-after-peak measure. If no point qualifies, the code inserts the final available sample time rather than a missing/censoring flag. This convention carries into post-processing.

4.4 Known processing limitations to communicate

Finite artifact rejection currently uses 'not any($x > A$) or any($x < -A$)', not the intended rejection of either positive or negative exceedances. Moreover, control and signal traces are filtered separately, while metadata repair handles recording-boundary exclusions rather than general artifact masks. Do not rely on finite artifact rejection as a validated paired rejection procedure. Blank/infinite threshold avoids finite-amplitude rejection; it does not remove the need for visual QC.

The preview calculates pre-/post-zero AUC of the pooled mean trace and a Welch t-test across its time samples. Its displayed asterisk is added unconditionally, not according to that p-value. Do not present this asterisk as evidence of a biological group difference. Time samples are not independent animals. Use an appropriate downstream analysis and inspect its actual results.

Legacy plot labels such as 'Shock Onset' are inherited text, not descriptions of a FED shock experiment. Save the figure with the verified event definition when presenting it.

5. FED3 post-processing: user workflow

Source: Post_processing/FED3_post_processing.py. This stage combines existing z-score event traces; it does not refit the control signal or renormalize raw fluorescence.

5.1 Select files and metadata

Click FED3 post-processing in the main window. Select compatible animal zScore .xlsx files. Output is saved beside the first selected file. Avoid selecting dFF/raw workbooks: extraction expects the z-score Time to baseline metadata row and may skip other signal types as invalid.

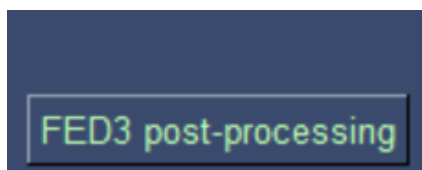


Figure 6. FED3 post-processing launcher.

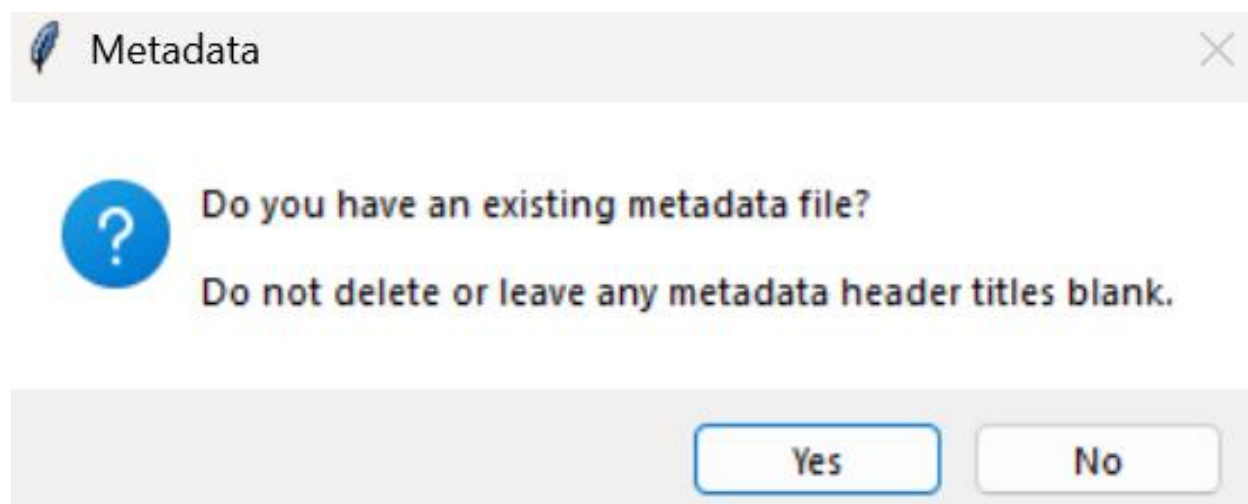
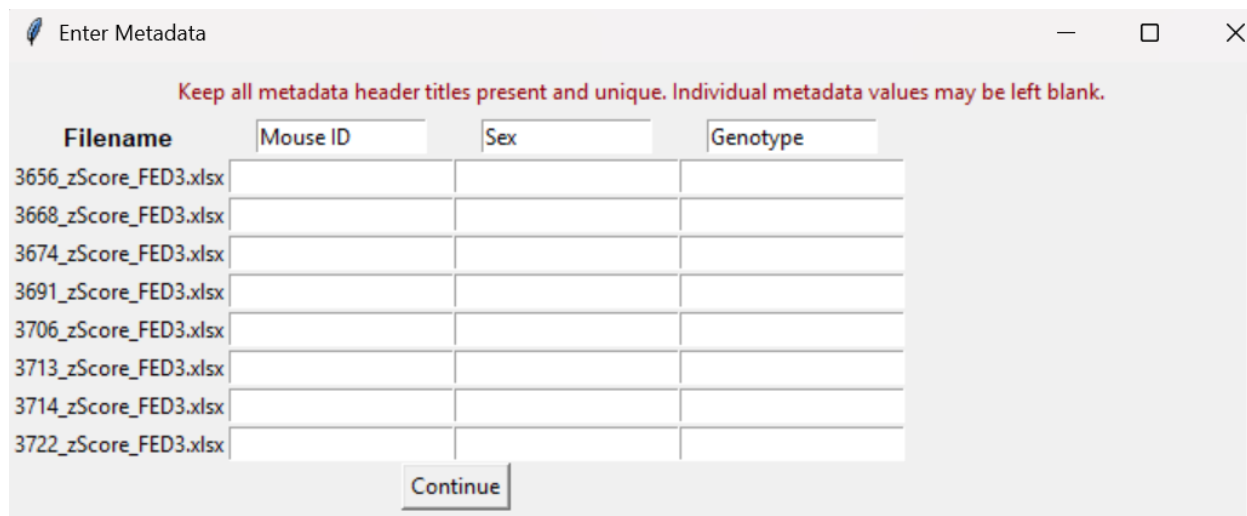


Figure 7. Choose whether to load existing metadata or enter it for this set of files.

If entering metadata, the default columns are Filename, Mouse ID, Sex and Genotype. The last three headers are editable, but their positions determine their roles: the first non-Filename column is the identifier, the second is the sex-like grouping, and the third is the genotype-like grouping. Keep all headers present, unique and nonblank. Do not reorder these roles in a reused metadata file.



Enter Metadata

Keep all metadata header titles present and unique. Individual metadata values may be left blank.

Filename	Mouse ID	Sex	Genotype
3656_zScore_FED3.xlsx			
3668_zScore_FED3.xlsx			
3674_zScore_FED3.xlsx			
3691_zScore_FED3.xlsx			
3706_zScore_FED3.xlsx			
3713_zScore_FED3.xlsx			
3714_zScore_FED3.xlsx			
3722_zScore_FED3.xlsx			

Continue

Figure 8. Metadata entry. Match filenames exactly and use consistent identifiers and group values.

The file is saved as FED3_FP_Metadata.xlsx. Reuse it only when filenames and metadata remain correct for the selected files; it is not automatically remapped to a new experiment. Rows are sorted by the third grouping column, then second grouping column, then numeric and textual mouse ID.

Use unique filenames and mouse IDs within a combined run. The code maps input basenames to paths and builds output columns keyed by mouse ID or mouse_eventN; repeated basenames or repeated IDs across sessions can overwrite dictionary entries or produce inconsistent metadata lengths. For longitudinal/repeated sessions, keep separate combined analyses until an explicit validated session-aware approach is available. Blank identifiers should not be used even though some entry fields allow blanks.

5.2 Select event categories

Only Left, Right, Pellet and Rewarded are offered. Overall and Extra event are not combined by this tool. Confirm that the same category means the same behaviour across every selected recording; the tool does not inspect the Arduino task or resolve a semantic mismatch.

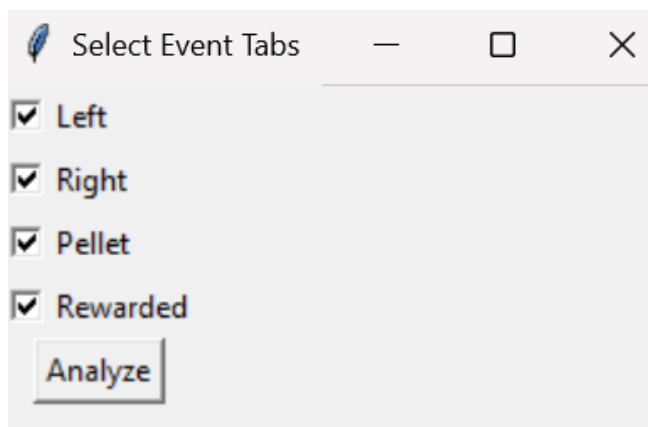


Figure 9. Select the normal FED3 categories to combine.

5.3 Configure progression figures

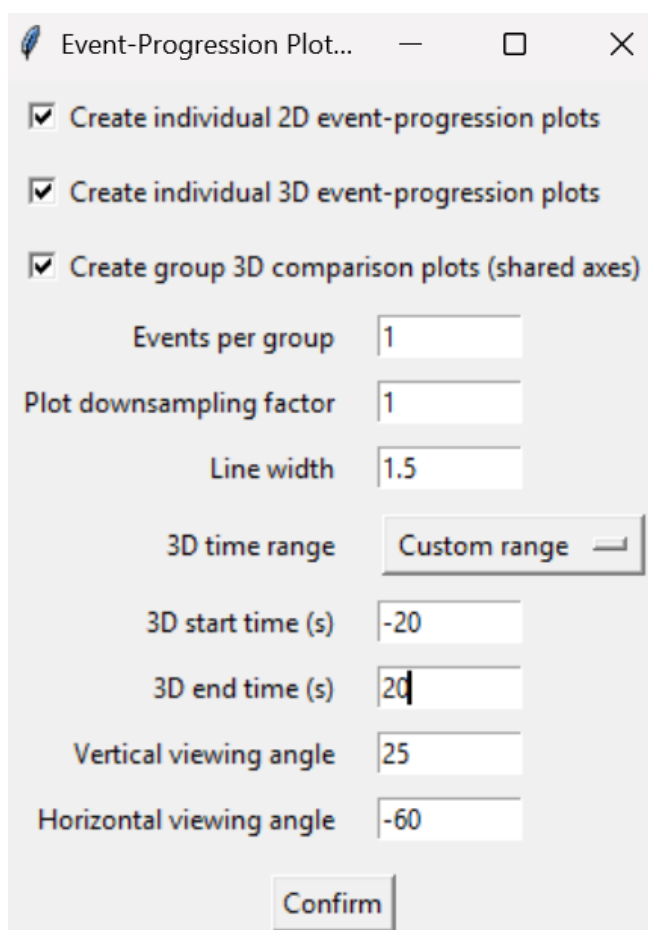


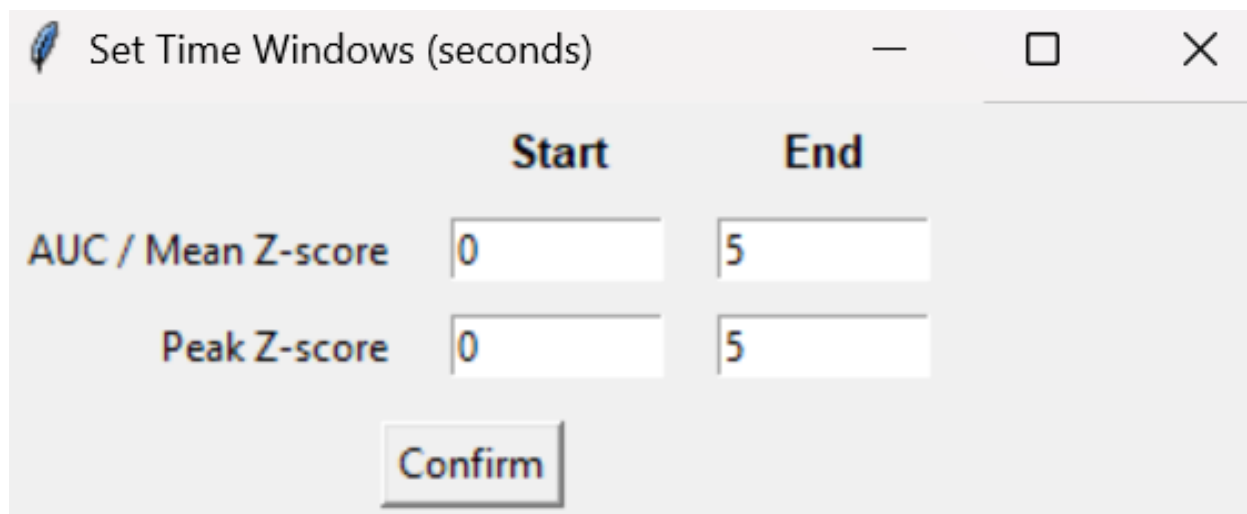
Figure 10. Event-progression options. Checked boxes and the custom range shown here are example choices.

Individual 2D plots, individual 3D plots and group 3D comparisons default to off in the current code. Events per group defaults to 1; plot downsampling factor to 1; line width to 1.5; 3D range to full available range; viewing angles to 25 and -60 degrees.

Events per group means consecutive event-index blocks, not groups of animals and not time bins. With a value of 2, events 1-2 form the first block, 3-4 the next, and so on. A final incomplete block is retained. A maximum of 20 blocks is displayed per individual 2D figure, with additional figures when needed.

Plot downsampling averages consecutive time samples for these progression plots only. It does not change the combined workbook, AUC/peak calculations or standard pooled overlays. The custom time range affects 3D plots only. Viewing angles, line width and colours change presentation, not calculations.

5.4 Set metric windows, colours and display



	Start	End
AUC / Mean Z-score	0	5
Peak Z-score	0	5

Confirm

Figure 11. Separate windows for AUC/Mean Z-score and Peak Z-score. Both default to 0-5 s.

These windows use actual peri-event seconds and include both endpoints when samples exist there. Choose them from the experimental question, not after inspecting which window gives the largest group difference. Require the same windows across a comparison. The code selects available samples, without interpolating exactly to the requested endpoints.

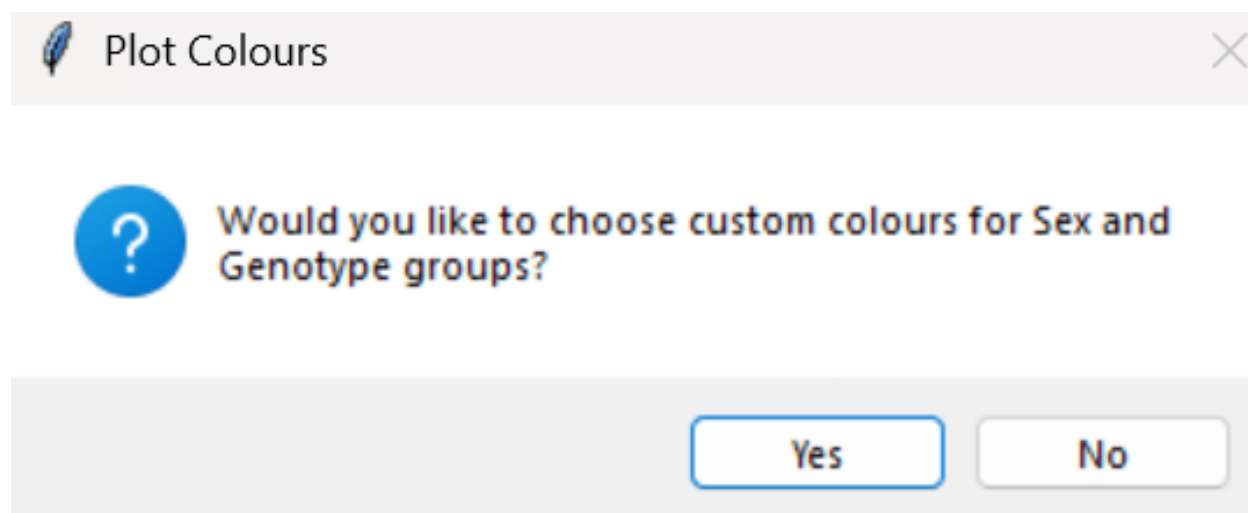


Figure 12. Optional group-colour selection.



Figure 13. Example colour picker. Colour changes have no effect on the statistical values.

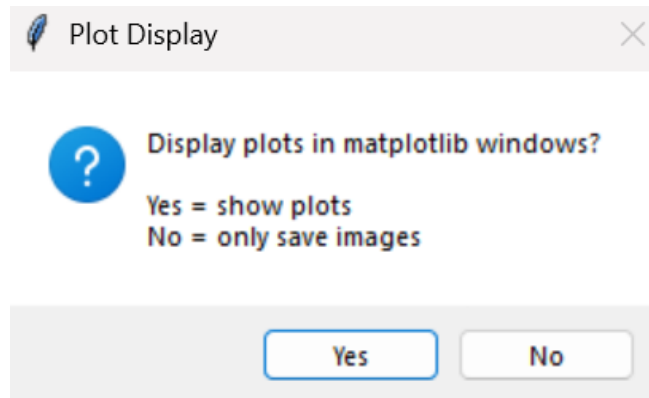


Figure 14. No saves figures without opening each Matplotlib window. The combined workbook is still created.

The combined file is FED3_FP_Combined.xlsx. Plots are saved under Plots, then the event category and plot grouping. Typical branches include Per_Mouse, grouping-column names, sex-by-genotype combinations, Subgroup_Comparisons and progression-plot folders. Inspect the console for skipped sheets/files before interpreting a completed run.

6. Post-processing mathematics and plots

Let $z_{mj}(t_k)$ denote sample k of event j for mouse/file m . A trial matrix has time samples as rows and events as columns. The extractor locates rows named Event note, Custom name and Time to baseline, retains exact matching event-note columns, and takes numeric times from the second worksheet column after Custom name.

6.1 Mean traces and SEM

Within a mouse, EventMeans is the pointwise arithmetic mean over events, ignoring NaNs:

```
mean_m(t) = nanmean_j z_mj(t)
```

Per-mouse plot shading is population SD across events divided by $\sqrt{\text{total number of event columns}}$. Standard group trace overlays horizontally concatenate all event matrices and use:

```
mean_group(t) = nanmean over all pooled event columns
```

```
SEM_group(t) = nanstd(population, pooled events) / sqrt(total pooled event columns)
```

Consequently a mouse contributing 100 events has ten times the weight of a mouse contributing 10 events in the pooled trace mean. This is not the mean of the animal means. With missing values, nanstd ignores missing entries but the denominator still uses total columns, not the number finite at each time. This is the exact code convention and should not be represented as a sample-SD animal-level SEM.

6.2 AUC and mean within a window

Let W contain the actual sample times satisfying $a \leq t \leq b$. For each event, the code computes signed trapezoidal AUC:

```
AUC_mj = sum_(k=1 to K-1) [(z_k + z_(k+1))/2] (t_(k+1)-t_k)
```

Units are z-score seconds. Negative portions subtract from positive portions; this is neither absolute area nor positive-only area. There is no additional baseline subtraction at this stage. np.trapz does not skip NaNs, so a NaN within a trapezoid can propagate to the AUC. A one-sample window yields zero trapezoidal area; use at least two valid samples.

```
MeanZ_window_mj = nanmean of z_mj(t) for t in W
```

Mean Z is an arithmetic sample average, not AUC divided by requested window length. For example, $t=[0,1,2]$ and $z=[0,2,0]$ give $\text{AUC}=2$ z-score seconds and $\text{MeanZ_window}=2/3$. Different endpoint weights explain why these are not interchangeable even on a regular grid.

6.3 Peak and peak time

Within the independently chosen peak window, `PeakValues` is the maximum finite/non-`NaN` z-score and `PeakValTime` is its event-relative sample time. Ties select the first occurrence. The maximum is not the largest absolute deflection: for an entirely negative trace it is the least negative value. An all-`NaN` event gives `NaN` peak and time. A peak at a window edge may indicate that the true peak lies outside the selected range.

6.4 TimeBaseline and cumulative metrics

`TimeBaseline` is copied from the initial processing header, not recalculated using the AUC/peak windows. It remains the first post-zero sample within `[-0.1,0.1]`, with the last sample inserted if no crossing occurs. '`CumTimeBaseline`' is a sum of these event-relative times, not session elapsed time or total pellet-retrieval time.

For any per-event sequence $v_1, \dots, v_J$, a cumulative sum and cumulative mean are:

$$C_j = \text{sum}_{(r \leq j, \text{ nonmissing})} v_r$$

$$CM_j = C_j / \text{count}_{(r \leq j, \text{ nonmissing})}$$

For peaks `[2,4,-1]`, the cumulative values are `[2,6,5]` and cumulative means `[2,3,1.667]`. A cumulative peak is a sum of peak heights, not an integral of fluorescence. Negative values can make a cumulative curve decrease.

Mice with fewer events are padded with `NaN` to the largest event count. Raw metric sheets remain blank after the last event, but cumulative exports treat padding as zero contribution, so the last cumulative value and mean persist across the padded tail. Apparent plateaus at late indices can therefore reflect no further events rather than biological stabilization. Leading missing data can yield zero cumulative values before the first valid event; a cumulative mean remains `NaN` until a valid count exists.

6.5 Event-index overlays and progression plots

Metric overlays align animals by event number, not absolute session time. At each event index the group curve is the mean of available mouse/file metric values; its SEM uses population SD divided by $\sqrt{\text{total mice/files in that grouping}}$. Late-event missingness changes the contributing sample size without changing that denominator. Cumulative transformations are applied within each mouse before the group average.

For progression plots with group size g , each mouse's block trace is the pointwise mean of its consecutive g events. The block SEM uses the count of non-`NaN` events at that time. Plot downsampling uses full d -sample blocks of both time and signal and drops an incomplete final time block. This full-block averaging is different from the $N-1$ slice in initial processing.

Individual 3D plots show peri-event time, event-block index and mean z-score. Group 3D plots first average each mouse's available events within a block, then average those mouse-block traces equally across contributing mice. They therefore use different weighting from the standard pooled-event 2D overlay. Shared axes aid visual comparison but do not constitute a statistical test.

Grouping plots may be produced by the third metadata field, by the second field, by their combinations, and by one field within levels of the other where multiple groups exist. These are descriptive subgroup displays, not automatic interaction tests.

7. FED3_FP_Combined.xlsx: every sheet explained

There is one Analysis Parameters sheet plus up to 13 sheets for each selected event category. E below means Left, Right, Pellet or Rewarded. If all four categories contain valid data, the expected total is 53 sheets. Missing/invalid categories can be skipped.

Sheet name	Rows and columns	Meaning
Analysis Parameters	One row per parameter	AUC/mean and peak windows, progression/display settings and colour choices. Not the full upstream normalization provenance.
E	Time rows; one column per event, named mouse_eventN	Individual z-score event traces. Main FiPhoPHA source for Individual events.
E EventMeans	Time rows; one column per mouse/file	Mean of that mouse's events at each time. FiPhoPHA source for Animal event means.
E PeakValues	Event-index rows; one column per mouse	Maximum z-score in the selected peak window.
E PeakValTime	Event-index rows; one column per mouse	Time of that peak relative to event onset, in seconds.
E CumPeakVal	Event-index rows; one column per mouse	Running sum of PeakValues.
E CumMeanPeakVal	Event-index rows; one column per mouse	Running mean of nonmissing PeakValues.
E TimeBaseline	Event-index rows; one column per mouse	Imported first post-zero crossing of [-0.1,0.1], or final time if absent.
E CumTimeBaseline	Event-index rows; one column per mouse	Running sum of TimeBaseline.
E CumMeanTimeBaseline	Event-index rows; one column per mouse	Running mean of nonmissing TimeBaseline.
E AUC	Event-index rows; one column per mouse	Signed trapezoidal area within the AUC window.
E CumAUC	Event-index rows; one column per mouse	Running sum of per-event AUC.
E CumMeanAUC	Event-index rows; one column per mouse	Running mean of nonmissing AUC.
E MeanZ_window	Event-index rows; one column per mouse	Arithmetic mean z-score within the AUC/mean window.

Each event-derived sheet has metadata rows for mouse ID, the third grouping field and the second grouping field, in that order. Headers can retain the user-entered names. The first spreadsheet column contains row labels/index; it is not another animal. The Time (s) or Event Number column identifies the horizontal domain.

Event numbering is restarted within each category and mouse in the combined workbook. The original TDT onset times, original global trial IDs and preceding-event notes are not carried into these combined sheets. Retain the first-stage workbooks for behavioural traceability.

All combined traces must use matching time vectors. The code compares them with `np.allclose(atol=1e-10)`, retaining NumPy's default relative tolerance; it does not resample mismatched files. Standardize initial processing settings and check sampling frequencies before combining.

8. FED3 FiPhoPHA: purpose and operation

8.1 Relationship to the published FiPhoPHA method

This local FED3-aware tool is based on the FiPhoPHA approach described by Vasilios (Vasili) Drakopoulos and colleagues, 'FiPhoPHA-A Fiber Photometry Python Package for Post Hoc Analysis', eNeuro 12(8), ENEURO.0221-25.2025, DOI: 10.1523/ENEURO.0221-25.2025. The published work provides the methodological context for bootstrap confidence intervals, permutation tests and consecutive-significance thresholds. Its temporal-threshold discussion links the choice to sensor kinetics and recording filters rather than treating one default as universally valid. [1]

The local Statistics/FED3_fiphopha.py file is a custom between-group implementation, labelled a draft in its source. It does not import the published fiphopha package and should not be described as an unchanged or fully validated reproduction. The equations below document this local code. In particular, its explicit difference-of-means bootstrap, window permutation procedure and endpoint adjustment must be reported as implemented.

8.2 Launch and load

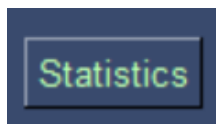


Figure 15. Open Statistics from the main photometry window.

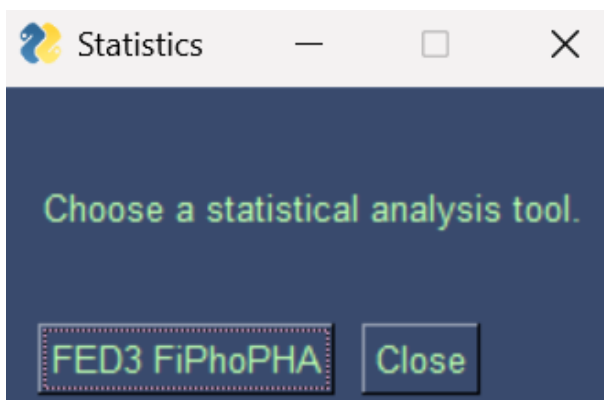


Figure 16. Select FED3 FiPhoPHA.

Choose FED3_FP_Combined.xlsx and click Load. The tool detects the available normal FED3 event sheets and metadata. It sets the displayed trace range from the first loaded event and normally splits pre-/post-event windows at zero. It retains downsampling factor 1 by default; the suggested factor near 1 Hz is informational, not automatically applied.

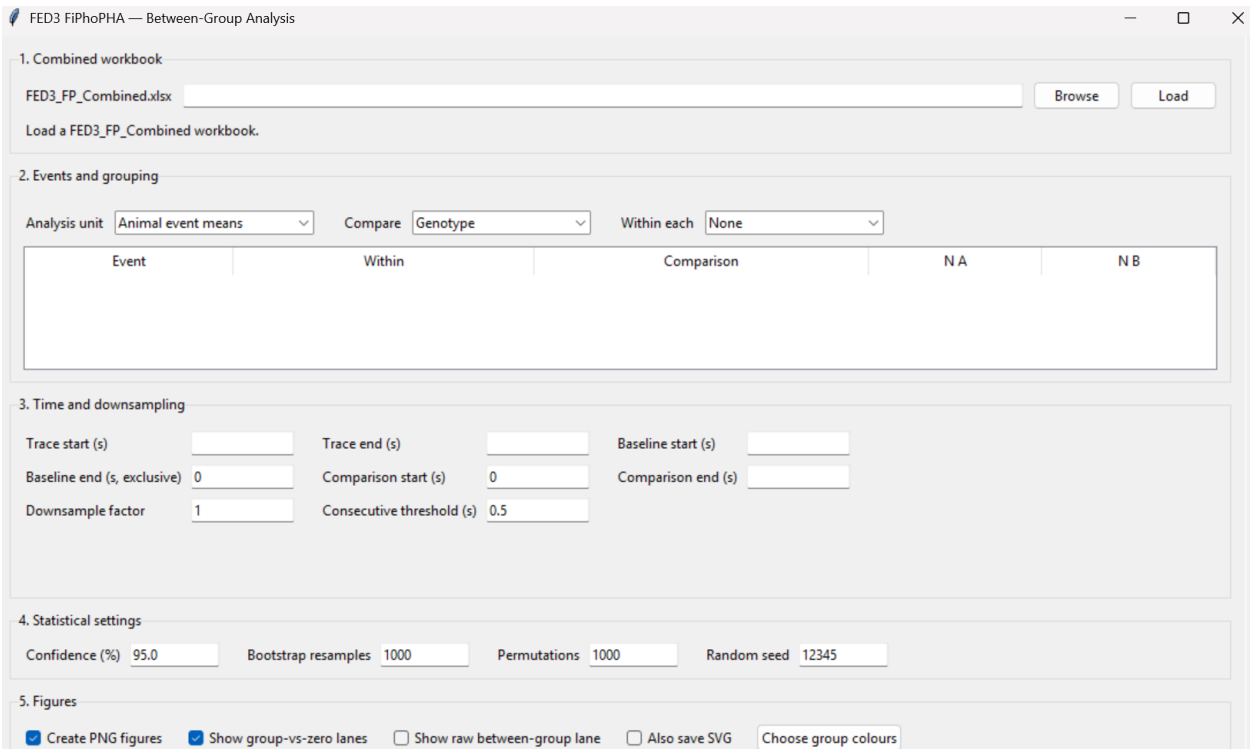


Figure 17. FED3-aware FiPhoPHA main window. The source default analysis unit is Animal event means.

It is a between-group tool only. 'Within each: Sex' means compare groups separately within each sex. It does not pair repeated measurements within an animal and does not perform a within-subject test.

8.3 Choose the analysis unit before interpreting N

Analysis unit	Source sheet	Each observation is	Consequence
Animal event means	E EventMeans	One mouse/file's mean event trace	Mice/files receive equal weight, assuming unique independent animals. This is the current default.
Individual events	E	One event trace	Mice with more events contribute more observations and weight. The code treats event columns as exchangeable independent observations.

2. Events and grouping

Events: ☒ Left ☒ Right ☒ Pellet ☒ Rewarded

Analysis unit: Compare: Within each:

Event	Within	Comparison

Figure 18. Example selecting Individual events and Genotype within Sex. This is an example selection, not a universal recommendation.

The draft's instruction 'We use Individual events' is not a generally valid statistical rule. Events from the same mouse are nested and can be correlated, especially with overlapping windows. This implementation does not perform mouse-clustered bootstrap/permutation or hierarchical modelling. For inference about independent animals, use a justified animal-level analysis or a separately validated hierarchical method. A large event N is not a large animal N.

Each comparison requires at least two observation columns per group. This software minimum does not establish adequate biological replication. Preview N is the number of selected columns, and can overstate usable N at particular times if values are missing. Completely empty columns are removed, but partly missing columns remain.

Choose Compare as a metadata field such as Genotype and optionally Within each as a different field such as Sex. The tool compares all nonblank levels pairwise within each selected stratum, sorted by label. A minus B follows that sorted pair order, not an automatically chosen control group. Confirm the comparison preview and Group Membership output.

8.4 Time, downsampling and statistical controls

3. Time and downsampling

Trace start (s)	-19.9017	Trace end (s)	60.0195	Baseline start (s)	-19.9017
Baseline end (s, exclusive)	0	Comparison start (s)	0	Comparison end (s)	60.0195
Downsample factor	1	Consecutive threshold (s)	0.5		

Source: 10.173 Hz (0.0983 s/sample). After factor 1: 10.173 Hz (0.0983 s/sample). The requested 0.5 s threshold becomes 5 point(s), effective duration $\approx$ 0.492 s.

Figure 19. Time/downsampling controls and the calculated effective consecutive threshold.

Control	Actual role
Trace start/end	Inclusive range retained before downsampling. Pointwise analyses cover this entire retained trace.
Baseline start/end	Window for a between-group permutation test; start inclusive, end exclusive. Does not re-zero or renormalize traces.
Comparison start/end	Second between-group permutation window; both endpoints inclusive. Does not restrict the pointwise bootstrap to this window.

Control	Actual role
Downsample factor	Full-block averaging of time and traces after trimming. Factor 1 retains source resolution; incomplete final block is dropped.
Consecutive threshold (s)	Minimum requested duration converted to an integer count of analysed samples; actual effective duration is reported.
Confidence (%)	Central percentile interval, default 95%. Not the probability a particular observed result is true.
Bootstrap resamples	Default 1000. Random resamples with replacement used for CIs, independently at each time point.
Permutations	Default 1000. Label shuffles used for each of the two window tests.
Random seed	Default 12345. Controls the reproducible pseudorandom sequence; it is not a threshold or sample size.

4. Statistical settings

Confidence (%) Bootstrap resamples Permutations Random seed

5. Figures

☒ Create PNG figures ☒ Show group-vs-zero lanes ☐ Show raw between-group lane ☐ Also save SVG

Figure 20. Statistical defaults are 95% confidence, 1000 resamples, 1000 permutations and seed 12345.

Loaded windows should be reviewed rather than left unchanged by habit. A window must contain analysed samples and fit inside the trace range. The GUI's threshold message explains how the selected duration is represented; it does not choose a biologically appropriate threshold for the sensor.

5. Figures

☒ Create PNG figures ☒ Show group-vs-zero lanes ☐ Show raw between-group lane ☐ Also save SVG

Figure 21. Plot controls: PNG, optional group-versus-zero lanes, optional raw significance and SVG export.

The default output workbook is FED3_FP_Combined_FiPhoPHA.xlsx. PNG figures are written to FED3_FiPhoPHA_Plots beside it; SVG is optional. Use a separate directory for each run because plot names may repeat across runs. Turning off a figure lane does not remove its results from the workbook.

9. FiPhoPHA mathematics, thresholds and seed

Source: Statistics/FED3_fiphopha.py. The unit selected in Section 8 defines the observation index below. The statistical assumptions must be assessed at that unit.

9.1 Means and plotted uncertainty

At each time t , group A and B means are the nanmeans of their observation columns. The reported difference is $D(t) = \text{mean_A}(t) - \text{mean_B}(t)$. A positive difference means group A has the higher numerical z-score.

Plotted SEM is population nanstd divided by $\sqrt{\text{total group columns}}$, matching the post-processing pooled overlay convention. It is not the bootstrap CI, and missing values do not reduce this SEM denominator. Two overlapping SEM bands do not by themselves determine the bootstrap result.

9.2 Pointwise bootstrap CIs

For each time point, the code removes non-finite values within each group. It draws B resamples of size n with replacement and calculates a mean for each. At confidence C , $q = (1-C)/2$, so the percentile endpoints are quantiles q and $1-q$; for 95%, these are 2.5% and 97.5%.

For the separate group-mean CIs, the current code multiplies both raw endpoints by $\sqrt{n/(n-1)}$ when $n > 1$. This scaling is about zero, not an expansion centred on the observed mean. Report this exact behaviour rather than describing it simply as a standard corrected bootstrap. It is not applied to the difference CI.

For the between-group difference CI, the groups are independently resampled at their own sizes for each resample b :

$$D_{b^*}(t) = \text{mean}(A_{b^*}(t)) - \text{mean}(B_{b^*}(t))$$

$$CI_D(t) = [\text{quantile}_q(D_{b^*}), \text{quantile}_{(1-q)}(D_{b^*})]$$

Raw Significant is 1 when this difference interval lies strictly above or strictly below zero. If either endpoint touches zero, the point is not significant. The code does not decide significance from whether the two separate group CIs overlap.

Group-versus-zero flags use each separate group CI in the same strict way and then apply consecutive filtering. These flags test numerical zero in the already normalized traces. They do not compare post-event values to a newly estimated baseline distribution in the FiPhoPHA window. With multiple group comparisons, the same group's bootstrap is recomputed in each pair using a different random sequence.

Resampling is performed independently at each time point, not as a single resample of complete trajectories across time. The pointwise result is not a simultaneous confidence band for the whole trace.

More resamples reduce Monte Carlo instability; they do not add animals or guarantee narrower intervals.

9.3 Consecutive-significance filtering

The analysed sampling interval is $dt = \text{median}(\text{diff}(\text{time}))$. For requested duration T :

$$K = \max(1, \text{round}(T/dt))$$

$$\text{Effective threshold} = K \times dt$$

$T=0$ explicitly gives $K=1$, which applies no consecutive filtering. Python rounding is used, including round-to-even at exact half values. This conversion uses rounding, not a ceiling, so the reported effective duration can be shorter or longer than requested.

The code finds uninterrupted runs of raw significant samples. A run survives only if its length is at least K ; when it survives, every sample in that run is retained. For $K=3$, $[0,1,1,0,1,1,1,0]$ becomes $[0,0,0,0,1,1,1,0]$. It does not only mark the third sample onward.

Filtering is based on the boolean significance flag, not the sign of $D(t)$. Adjacent positive and negative significant differences can be retained as one run if no nonsignificant sample separates them. The interval's reported direction then follows its average difference, so inspect pointwise values if direction changes matter.

Approximate analysed sampling	Requested T	K	Effective duration
10 Hz ($dt=0.1$ s)	0.5 s	5	0.5 s
5 Hz ($dt=0.2$ s)	0.5 s	2 at an exact 2.5 ratio	0.4 s
1 Hz ($dt=1$ s)	0.5 s	1	1 s; isolated points are not removed
Screenshot source about 10.173 Hz	0.5 s, factor 1	5	About 0.492 s

Choose the duration based on sensor kinetics, acquisition filters and the scientific question, then verify that the sampling resolution can represent it. The software warning at $K=1$ is important: one point does not filter isolated detections. Do not increase downsampling solely to speed analysis without reconsidering temporal resolution.

Consecutive filtering addresses short isolated crossings, but this local implementation does not run a cluster-permutation test or apply a global multiple-comparison correction across all times, events, pairs and strata. It does not guarantee a specified family-wise error rate for arbitrary data. Define comparisons and report the limitation.

9.4 Window-level permutation tests

Baseline and comparison windows each produce a separate two-sided independent-group test. The code first averages time samples within the window for each observation:

```
u_Ai = nanmean_(t in W) A_i(t); u_Bj = nanmean_(t in W) B_j(t)
D_obs = mean_i(u_Ai) - mean_j(u_Bj)
```

It pools these scalar observation means, randomly shuffles them, splits at the original group sizes and calculates D_{perm} . If E of P shuffled differences satisfy $|D_{\text{perm}}| \geq |D_{\text{obs}}|$, then:

$$p = (E + 1) / (P + 1)$$

For $P=1000$, the smallest possible reported p is $1/1001$, about 0.001 , not zero. These p -values test group differences in window means. They are not tests of a within-group pre-versus-post change, an AUC contrast or a time-by-group interaction. No consecutive threshold is applied to these two scalar tests.

The permutation assumes observations can be exchanged between groups under the null. Individual events from the same mouse may violate that assumption. Repeated measures and paired conditions are not modelled here. Choosing Within each only restricts which observations enter each independent comparison.

9.5 Random seed and reproducibility

The seed initializes NumPy's pseudorandom generator. For successive event/stratum/group-pair comparisons, the code uses `base_seed + comparison_counter`, beginning at zero. Within a comparison the same generator is consumed by group bootstraps, difference bootstraps and then both permutation tests.

The same input, settings, selected events, ordering, code and numerical-library versions should reproduce the result. Changing the seed, sample order, resample count, trace range or earlier comparisons can change the random draws and slightly change CI boundaries/ p -values. Adding a comparison can shift seeds assigned to later comparisons. The base seed is saved, but per-comparison seeds are not separate output columns; archive selections and code as well.

Seed 12345 is a reproducibility convention, not a preferred biological result. Do not search seeds for significance. If a conclusion depends on small Monte Carlo fluctuations near a cutoff, report that instability and assess it transparently rather than selecting a favourable run.

10. FiPhoPHA results: every Excel tab

This workbook contains seven sheets. It does not overwrite or append statistical results to the original combined workbook. Column sets with group names can vary across comparisons, leaving unrelated columns blank in the stacked result tables.

10.1 Analysis Summary

One row per event, stratum and group pair. It records Event, Analysis Unit, Group Field, Within Field/Value, Comparison, group N, baseline mean difference and p-value, comparison mean difference and p-value, significant downsampled point count, K and effective duration. N is observation columns, not necessarily animals. Significant Downsampled Points counts all retained points across runs; it is not the number of significant intervals.

The two p-values correspond to the two window tests. Pointwise bootstrap results can show a brief difference even when a whole-window mean test is nonsignificant, and the converse can occur. These are different questions, not necessarily contradictory results.

10.2 Significant Intervals

One row per contiguous run retained by the between-group threshold. Columns identify the comparison, first/last sample time, approximate duration, average A-B difference and direction. Duration is $\text{number_of_samples} \times \text{dt}$, whereas End minus Start equals $(\text{number_of_samples}-1) \times \text{dt}$. Thus a single retained sample can have a nonzero approximate duration. No retained intervals can leave this sheet empty, without a populated header table in the current writer.

Direction names the group with the higher interval-average value, not the direction of change relative to its own baseline. Intervals are computed over the full trace range, so pre-event intervals are possible.

10.3 Pointwise Results

One row per comparison and analysed time. Includes both means, both SEMs, group CI endpoints, Difference (A-B), difference CI endpoints, Raw Significant and Thresholded Significant. Flags are 0/1. This is the detailed table to inspect when verifying sign, isolated crossings or the effect of the consecutive filter. There is no pointwise permutation p-value column.

10.4 Significance Map

A compact long-format table of Event, Analysis Unit, grouping/stratum, comparison, Time (s), Raw Significant and Thresholded Significant. It is not a matrix with mice as columns. Use it for plotting timing, not as continuous fluorescence data.

10.5 Significance From Zero

One row per event/stratum/group/time combination after identical rows are dropped. Significant From Zero is the thresholded separate-group CI flag. The test is against zero, not against the other group and not a newly fitted baseline. Because group CIs are recalculated in each pair, more than two groups can produce differing duplicate-key flags from different Monte Carlo draws; the sheet has no Comparison column to distinguish those cases. Inspect pairwise pointwise CIs when this occurs.

10.6 Group Membership

Lists Event, Analysis Unit, Observation and metadata values from the loaded datasets. Use it to check IDs, group labels and what a column represents. It inventories loaded observations, including ones that might not enter a pair because their compare/stratification labels are blank or a stratum has fewer than two groups. It is not a per-pair final inclusion log.

10.7 Analysis Settings

Records source path, between-group analysis type, confidence fraction (0.95 means 95%), resamples, permutations, requested threshold, downsample factor, analysed interval, trace/window bounds, seed, p-value convention and colours. The reported downsampled interval comes from the first loaded dataset. Original sampling interval is a descriptive note rather than a numeric value for every event. Preserve the source workbook and upstream processing settings alongside this sheet.

10.8 Reading the figures

Each comparison figure has upper significance lanes and lower mean plus/minus SEM traces. The main upper lane is thresholded between-group significance; optional lanes show raw between-group flags and each group versus zero. These lines represent time intervals, not effect sizes. They are drawn over sample support, approximately half a sampling interval beyond the centres at run edges.

The lower plot includes a zero line, event onset, lightly shaded baseline/comparison windows and their permutation p-values. Its shaded bands are SEM, not the bootstrap CIs used to generate upper lanes. A plot colour or lane option never changes the underlying inference.

11. Quality control, interpretation, troubleshooting and sources

11.1 Before processing and combination

- Verify actual event mappings with the FED log and TDT circuit; do not infer them solely from 'Left' or 'Pellet'.
- Confirm control/signal channels belong to the same animal and setup. A prefilled option is a candidate, not a validation.
- Uncheck unused Extra event. Check for duplicated sources and cross-setup event selections.
- Keep identical normalization, time grids and behavioural definitions across the files being combined.
- Inspect traces for movement artefacts, clipping, non-finite values and overlapping behaviours. Software output is not automatic quality approval.
- Use unique IDs and source basenames; archive one combined run per intended independent dataset.

11.2 Common symptoms

Symptom	Check first
Left counts differ from FED total Left_Poke_Count	Rewarded pokes may be in Arwd; dispensing/with-pellet pokes may lack the custom BNC; boundary windows may be excluded.
Missing/empty category	No acquired epocs, wrong mapping, category unticked, invalid header structure or skipped file. Check console messages.
No data after processing	t-range may exceed recording coverage. Reduce it only if scientifically appropriate and apply consistently.
Post-processing skips a workbook	It expects zScore structure, exact event notes, Custom name and Time to baseline rows.
Time-vector error	Initial windows/sampling differ. The tool does not interpolate them into agreement.
Unexpected repeated event columns	The same TDT stream may have been assigned to multiple GUI categories.
Late cumulative curves plateau	Padding after the last event can carry the last cumulative value forward.
FiPhoPHA reports too few observations	Check analysis unit, missing groups, metadata spelling and subgroup sizes.
Threshold becomes one point	Downsampling is too coarse for the requested duration; isolated significant points will survive.

Symptom	Check first
Results change with seed	Monte Carlo draws changed; investigate borderline conclusions rather than selecting a favourable seed.
Plot mean differs between tools	Check pooled events versus animal means, downsampling, range and group 3D weighting.

11.3 Reporting results

Report the physical behaviour and source-store mapping, not just the Excel label. State sensor, acquisition filters/sampling, preprocessing baseline and nominal N, post-processing windows, number of animals and events, FiPhoPHA analysis unit, compare/stratification fields, trace range, confidence, resamples, permutations, seed, and both requested and effective consecutive thresholds.

Specify whether an effect is a window-mean permutation result, a thresholded pointwise difference CI or a group-versus-zero flag. Report A-B direction and biological replication separately from event count. Neither descriptive progression plots nor cumulative summaries independently establish statistical significance.

Missing event types and failed trial extraction should be reported rather than silently interpreted as zero biological response. An event-free animal is not the same as an animal with measured zero-valued event traces.

11.4 Literature and software references

[1] Drakopoulos, V., and colleagues (2025). FiPhoPHA-A Fiber Photometry Python Package for Post Hoc Analysis. *eNeuro*, 12(8), ENEURO.0221-25.2025. <https://doi.org/10.1523/ENEURO.0221-25.2025> . Published package repository: <https://github.com/VasiliDrakopoulos/fiphopha> . Cite the paper when describing the methodological basis; additionally identify the local FED3 adaptation and revision used.

[2] Kravitz Lab Devices, FED3 library. https://github.com/KravitzLabDevices/FED3_library . The supplied sketch includes FED3.h; archive the library version actually compiled for the experiment. Public library source was used to clarify BNC timing, but it does not prove every historical/custom build behaves identically.

[3] The local FibPhoEpocAveraging.py header credits TDT, David Root and the Morales Lab, with a workflow inspired by Barker and colleagues (2017). Preserve those original software acknowledgements when distributing the GUI and distinguish this preprocessing lineage from the FiPhoPHA statistical lineage.

12. Day-to-day quick reference

Use this checklist after agreeing your analysis settings. It summarises the routine workflow; the earlier sections explain the calculations and limitations.

Before starting: Keep raw files unchanged, use a separate output folder for each run, and confirm the event mapping for the session mode and setup.

12.1 Process each recording

- Open the GUI. Select the complete TDT block folder, export folder, setup, matching camera and FED3 analysis.
- Verify the ISOS/control and signal streams. Select Normal task for the usual FED3 workflow.
- Choose the event categories and their verified TDT stores. Leave Extra event unticked unless needed.
- Check the peri-event range and baseline. The two t-range entries are start time and duration: -20 and 80 means approximately -20 to +60 seconds.
- Enable z-score export for post-processing. Run, inspect the output and retain the settings. Repeat for other animals/setups using compatible settings.

Output: Per-recording Excel workbooks. Overall combines selected events; Left, Right, Pellet and Rewarded contain events from their assigned stores.

Mapping reminder: A/B denotes the setup, not active/inactive. In the reviewed normal-PR example, lft = unrewarded active, rgt = inactive, rwd = rewarded active and plt = pellet retrieval. Do not assume this mapping holds for, let's say, FR1R or other modes. For normal PR sessions using the referenced ClassicFED Arduino code and our Andrews Lab TDT setup, A/Blft records unrewarded active left pokes, A/Brgt records inactive right pokes, A/Brwd records reward-triggering left pokes, and A/Bplt records pellet retrieval. These definitions are not universal across the Arduino modes. For example, in FR1R the physical right poke is active, and its BNC signal can be recorded as A/Blft and therefore appear in the GUI's Left tab. The GUI sorts events according to the selected TDT store. It does not determine which physical port was poked or whether that poke was rewarded.

12.2 Combine recordings: FED3 post-processing

- Click FED3 post-processing and select compatible animal z-score workbooks. Use unique filenames and mouse IDs within the run.
- Enter metadata or load a file that matches the selected recordings. Choose Left, Right, Pellet and/or Rewarded.
- Optionally select 2D/3D progression plots. Events per group combines consecutive events; plot downsampling affects progression figures, not the exported event traces.
- Set the AUC/mean and peak windows. Choose colours. Select No to save plots without opening every plot window.

Output: FED3_FP_Combined.xlsx, FED3_FP_Metadata.xlsx and a Plots folder beside the first selected file. Post-processing summarises existing z-scores; it does not renormalize them.

12.3 Compare groups: FED3 FiPhoPHA

- Open Statistics → FED3 FiPhoPHA. Load FED3_FP_Combined.xlsx and select the event categories.
- Choose the analysis unit. Animal event means uses one mean trace per mouse/file; Individual events uses each event as an observation. Events from the same animal are not independent animals.
- Choose Compare (for example, Genotype) and optionally Within each (for example, Sex). This compares groups separately within each subgroup; it is not a paired within-animal test.
- Review the trace range, baseline/comparison windows, downsampling and consecutive threshold. Check the displayed effective threshold, not only the requested duration.
- Review confidence, bootstrap resamples, permutations and random seed. Keep the agreed settings consistent; defaults are starting values, not universal recommendations.
- Choose figure options/colours, run the analysis and save the results in a distinct output location. Check group membership and sample counts before interpreting significance.

Output: A separate FiPhoPHA results workbook and, if enabled, figures. Start with Analysis Summary for window-test results, Significant Intervals for retained time periods, and Analysis Settings for the run settings.

12.4 Terms at a glance

Z-score: control-corrected fluorescence expressed relative to the chosen normalization baseline.

AUC / mean / peak: signed area, average value, or highest value within the selected metric window.

Event progression: responses arranged from first to last event. Events per group averages consecutive event blocks.

Downsampling: averaging consecutive time samples to reduce temporal resolution. FiPhoPHA downsampling affects its statistical analysis.

Consecutive threshold: the minimum run of significant samples to retain, converted from seconds at the analysed sampling rate. Check the effective duration.

Bootstrap / permutation: resampling for confidence intervals / shuffling group labels to test differences in window means.

Random seed: controls reproducible random draws. Keep it with the settings; do not change it to seek significance.

Final check: Verify event meanings, compatible settings and the analysis unit. A window-test p-value and a significant time interval answer different questions. Use Sections 7 and 10 for the full Excel-tab reference.